

v4 defringe · one mask per frame

Coded in batch_defringe/process_v4.py. Does not overwrite defringe_v22 or v3. notes/V4_PIPELINE.md

1. Stack hints (not locked q) · Experiment.xml catalog · shutter FOV-std cliff · shutter-learn qs
Each frame re-seeds. Last-frame q is not tracked. Empty mask allowed.

- 2a. Collect lines (this order)
 - catalog q · shutter-learn q
 - linescan center peak (if congruent)
 - same-trace edge / segment qs (chirp)
 - leftover FFT is later, after a mask exists

- 2b. Linescan congruence
 - H, V, both diagonals always
 - One (qy, qx): fx / fy / tilted — or none
 - none → skip linescan pieces; still catalog, shutter-learn, leftover FFT

3. Add lines into ONE mask at low α ($0.28 \approx \frac{1}{3}$ of pack_D full 0.85)
Each trial: FFT notch → predicted (IFFT of taken-out energy) and removed (raw – cleaned).
Keep only if removed still looks like fringe and tracks predicted. Live: undo if cells. Shutter: push.
Then leftover FFT on cleaned; try those lines at 0.28 too (dubious extras, not a second family).

4. Raise α collectively on the whole accepted set: $0.55 \rightarrow 0.85$ (shutter also 1.00). Stop at first reject.
One shared max_alpha — not per-line. Peak pieces: $\text{att} = \alpha \times \text{peak_mask}$. Ridge pieces: pack_D local excess with that same α , and a per-line spectral gate (gate=0 skips that ridge). Combined by max.
Live search ± 10 bins around this frame's own hint; shutter ± 2 . No stack q-tracker.

5. Write · dest --out: cleaned TIFF, slim overview.pdf, per_frame.csv, families.json, mask_recipe.json (rung story), mask_patterns.npz (sparse final + per-rung FFT masks).
E:\Rasmus-Guillermo is write-guarded. Removed/mean TIFFs not stored on dest; rebuild from npz + raw.

Congruence · one (qy, qx) or none

All four cuts always run: several horizontal rows, several vertical columns, both diagonals.
A fringe is one plane-wave, so the cuts must describe the same 2-D frequency — allowing for angle.

Hypotheses (scored against measured diagonal periods):

- f_x — vertical stripes (intensity varies along x). q_x from the HORIZONTAL cut; $q_y = 0$.
The vertical cut can be quiet; that is expected, not a failure.
- f_y — horizontal bands. q_y from the VERTICAL cut; $q_x = 0$.
- tilted — both q_x and q_y . Diagonals must match the predicted period at (q_y, q_x) .

Score = mean relative error of predicted vs measured main/anti diagonal periods.

Winner = lowest score. If no hypothesis, or best score $> 0.25 \rightarrow$ congruence is none.

none is honest on shutter 2-D ridges (not a 1-D stripe). Then skip linescan center/edges.

This 'agree' is not the predicted $\leftrightarrow$ removed agree later. It is only 'do the four cuts describe one orientation?'

α · what 0.28 is · one mask, not per-family α

α is the attenuation cap on the FFT support (how hard we notch), not a spatial pixel mask.

Rungs (chosen, not measured from the data):

live 0.28 → 0.55 → 0.85 (0.28 $\approx$ $\frac{1}{3}$ of pack_D full 0.85)

shutter 0.28 → 0.55 → 0.85 → 1.00

Adding a line: fold it into the same mask, apply the current set at 0.28. If the trial fails, that line is not kept. After leftover lines are considered, α is raised on the WHOLE accepted set together. Families do not get separate α schedules.

v2.2 / pack_D DID use per-family gate and eff_max_alpha (per_frame.csv family{i}_*).

v4 dropped independent families on purpose: leftover / chirp / harmonics are extra support of the same fringe, not a second cleaner. Ridge lines still have pack_D gate from this frame's spectral strength (gate=0 skips that line). Peak (linescan) pieces have no gate — only shared α . Open: per-line α later, so a dubious leftover is not pushed as hard as the core. Not implemented.

Predicted vs removed · RMSE / L2 / agree

predicted = IFFT of the FFT energy the current mask attenuates (what we intended to take out).

removed = raw – cleaned (what actually left the image).

They should look like the same striped fringe. Cells in removed that predicted does not have means those bins held biology — undo the last step only.

All scalars are over every pixel of that one frame (not tiles, not a mean of row-RMSs):

$\text{RMS}(\text{pred}) = \sqrt{\text{mean}(\text{pred}^2)}$

$\text{RMS}(\text{removed}) = \sqrt{\text{mean}(\text{removed}^2)}$

$\text{RMSE} = \sqrt{\text{mean}(\text{removed} - \text{pred})^2}$

That RMSE is the L2 (Euclidean / sum-of-squares) distance between the two images.

L1 would have used $|\text{removed} - \text{pred}|$ instead.

$\text{agree} = 1 - \text{RMSE} / (\text{RMS}(\text{pred}) + \text{RMS}(\text{removed}))$

1 = identical images. Live keep if $\text{agree} \geq 0.40$ and removed still looks like fringe (coverage / even, and blob-or-ridges). Reject if $\text{agree} < 0.40$, removed RMS > 0.5, and the increment looks like cells. Shutter skips the biology brake.

Removed RMS vs 0 is a heaviness trace in the PDF/CSV, not this success test.

(fx, fy) between frames · no stack tracker

v4 does not lock q across time. Each frame re-collects catalog / shutter-learn / linescan.
There is no $\pm N$ bin bound on how much the mask location may jump from frame n to n+1.

The only snap is INSIDE a frame, and only for ridge pieces:

live search ± 10 bins around that frame's own hint (TRACK_SEARCH)

shutter search ± 2 bins

Q_CLUSTER_TOL = 3 bins only merges duplicate lines on the SAME frame (not tracking).

Peak pieces use this frame's linescan (qy, qx) with no inter-frame bound.

Do not carry forward from v2.2/v3: stack q-tracker (± 10 / apply ± 2) as the product.

Do not union-apply frozen pack_D families. Do not treat edge-q as family 2.

Do not IFFT whole FFT rows/columns. Do not overwrite defringe_v22.

What 'better' looks like · dest readout

At each rung, predicted should get more complete and still stripe-like / conjugate-symmetric. Removed should track it. Leftover should lose fringe, not cells. Undo the increment if not.

overview.pdf (slim dest run): cover, shutter cliff, traces (RMS / n lines / α / agree / brake), means of raw / cleaned / removed / predicted. Full rung stories live in mask_recipe.json + mask_patterns.npz (sparse applied FFT after every kept and undone step).

per_frame.csv: frame, role, n_lines, max_alpha, removed_rms, removed_mean_abs, removed_p99_abs, agree, brake, empty, core_only, seed_winner, qx, qy, accepted, n_history, n_undone.

Dest batch · USB keepalive · progress

```
python -m batch_defringe.process_v4 --root E:\Rasmus-Guillermo\ECF1 --out F:\CollectedData
```

E:\Rasmus-Guillermo is write-guarded. Dest gets folder structure, Experiment.xml, cleaned TIFF, slim PDF, per_frame.csv, families.json, mask_recipe.json, mask_patterns.npz. Resume skips complete dest folders.

USB: keepalive every 20s (read source, write dest .defringe_v4_runs/keepalive.txt). Retry a stack 3x on Permission denied / 0 written / vanished drive (wait up to 3 min). Stop after 3 consecutive drive failures. The script does not change Windows power options. USB selective suspend Off is still the stronger OS fix.

Progress: console ETA, plus .defringe_progress.txt in the repo and F:\CollectedData\defringe_v4_runs\<utc>\progress.txt